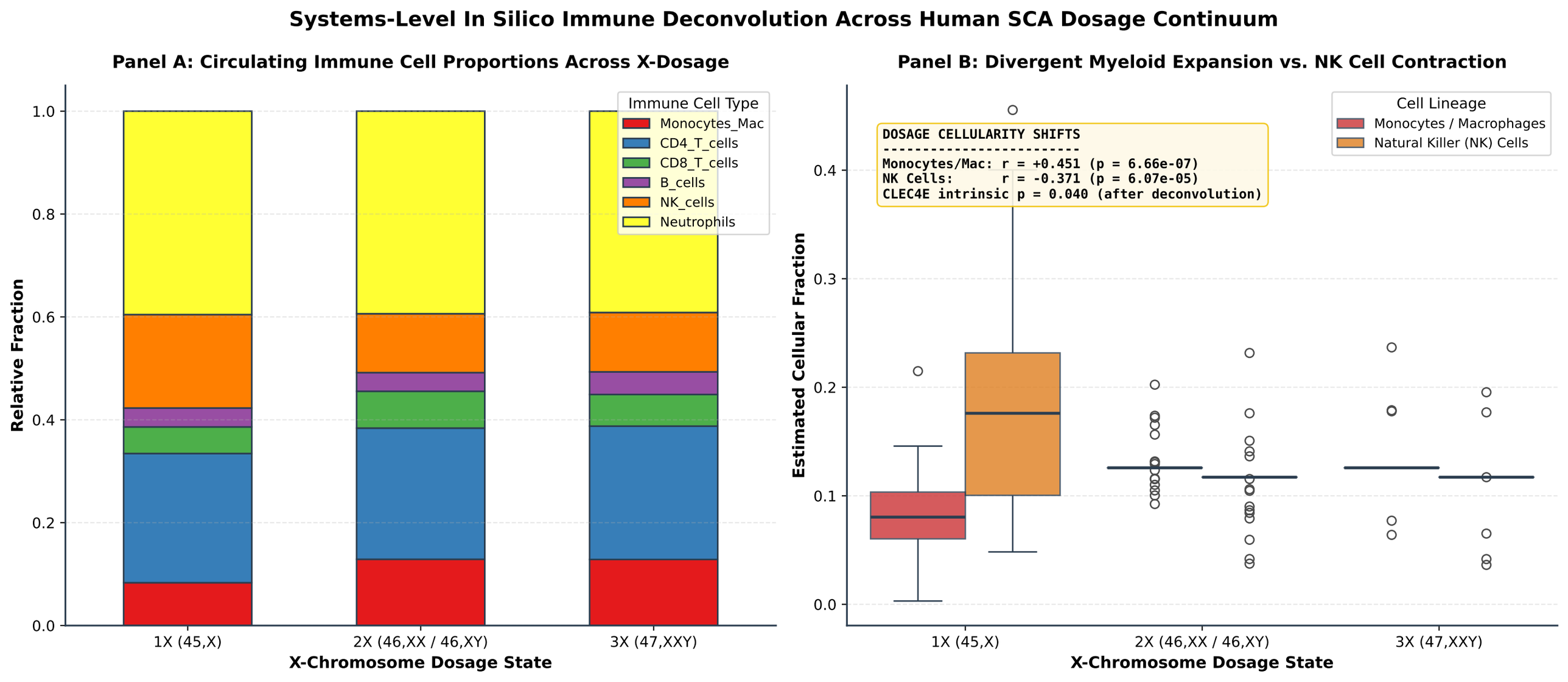
Systems-Level In Silico Immune Deconvolution Across Human SCA Dosage Continuum
Panel A: Circulating Immune Cell Proportions Across X-Dosage
Immune Cell Type
Monocytes_Mac
CD4_T_cells
CD8_T_cells
B_cells
NK_cells
Neutrophils
Relative Fraction
1.0
0.8
0.6
0.4
0.2
0.0
1X (45,X)
2X (46,XX / 46,XY)
3X (47,XXY)
X-Chromosome Dosage State
Panel B: Divergent Myeloid Expansion vs. NK Cell Contraction
DOSAGE CELLULARITY SHIFTS
Monocytes/Mac: r = +0.451 (p = 6.66e-07)
NK Cells: r = -0.371 (p = 6.07e-05)
CLEC4E intrinsic p = 0.040 (after deconvolution)
Cell Lineage
Monocytes / Macrophages
Natural Killer (NK) Cells
Estimated Cellular Fraction
0.4
0.3
0.2
0.1
0.0
1X (45,X)
2X (46,XX / 46,XY)
3X (47,XXY)
X-Chromosome Dosage State